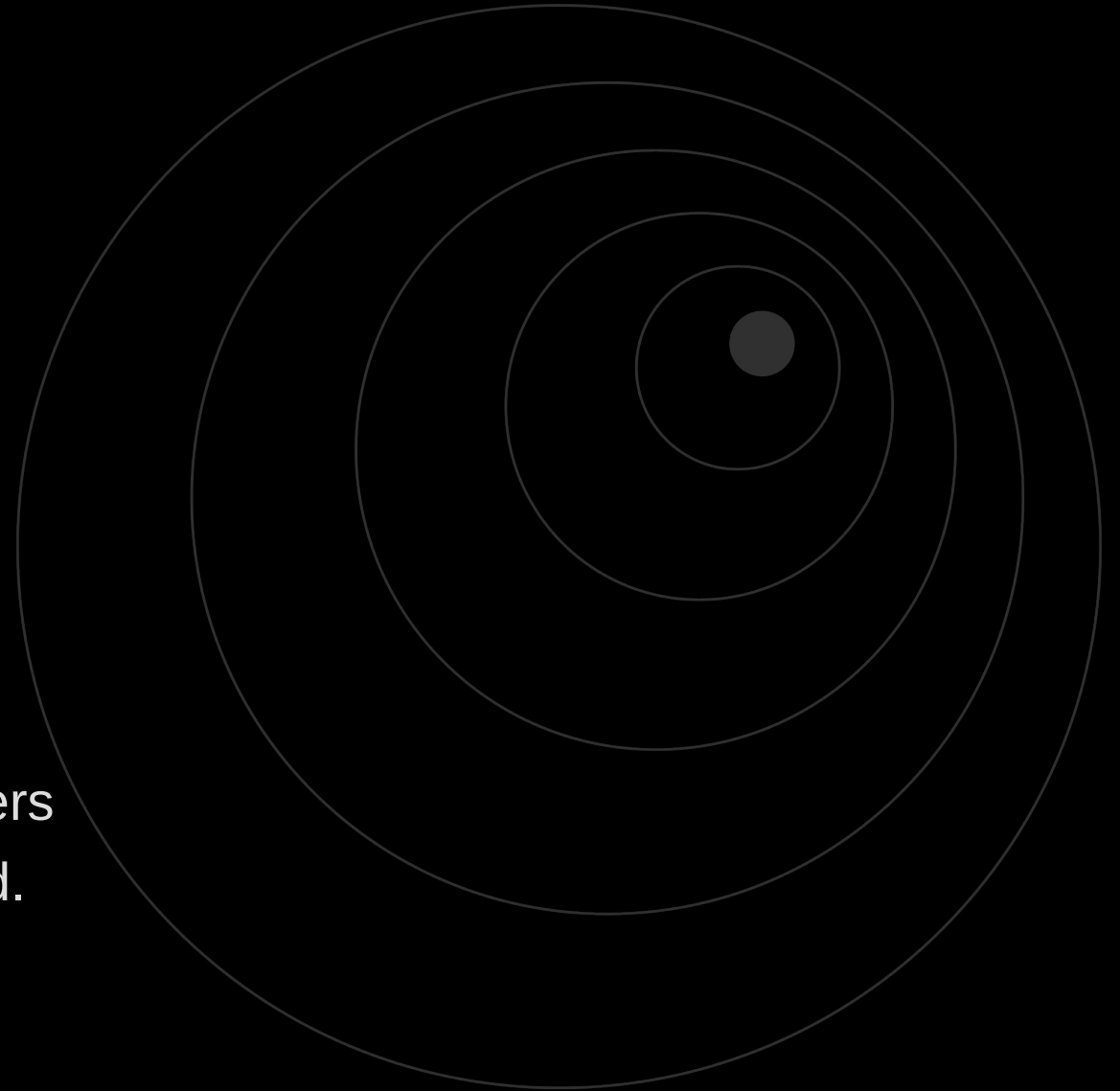


# R ROTO

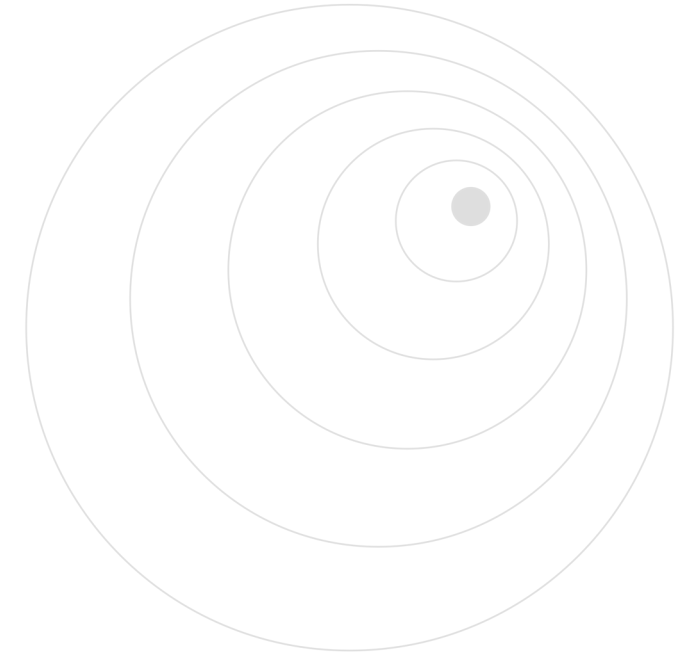
Aerial data for quarries. Weekly stockpile numbers instead of a once-a-month surveyor walk-around.



PROBLEM

# A quarry is blind between surveys

The staff surveyor walks the yard with a total station once a month. It takes half a day. Everything between those visits, short-loading, shrinkage, plain theft, piles up unseen until the next survey.



# 1x

per month is how often most yards get measured

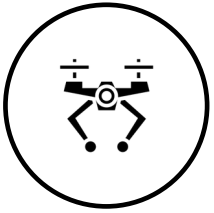
# 400–700K

KZT monthly salary of a mine surveyor (estimate, hh.kz vacancies)

# +15.9%

construction growth in Kazakhstan in 2025, quarries feed it on monthly numbers

# Drone scan in, volume report out

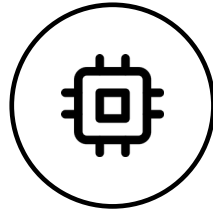


## Fly

A 249 g drone flies an automated route over the stockyard. Twenty minutes per yard.

## under 5%

target error vs manual survey, proven on a reference pile in week one



## Process

Photogrammetry pipeline on open-source WebODM builds a 3D model and computes each pile's volume.

## under 500K KZT

one-off hardware cost, bought in Almaty within 3 days



## Report

Deltas against the previous scan, full stock dynamics. A subscription, 2 scans a month.

## zero permits

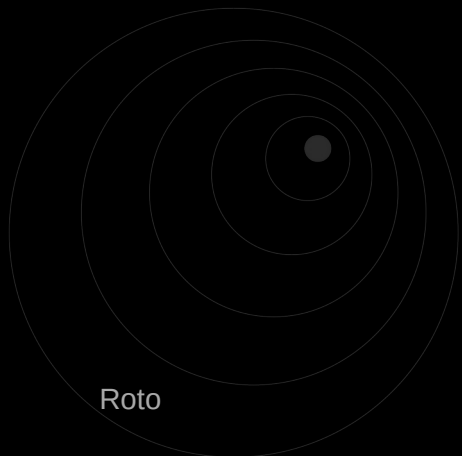
under the 250 g threshold, outside populated areas, open category

# 28

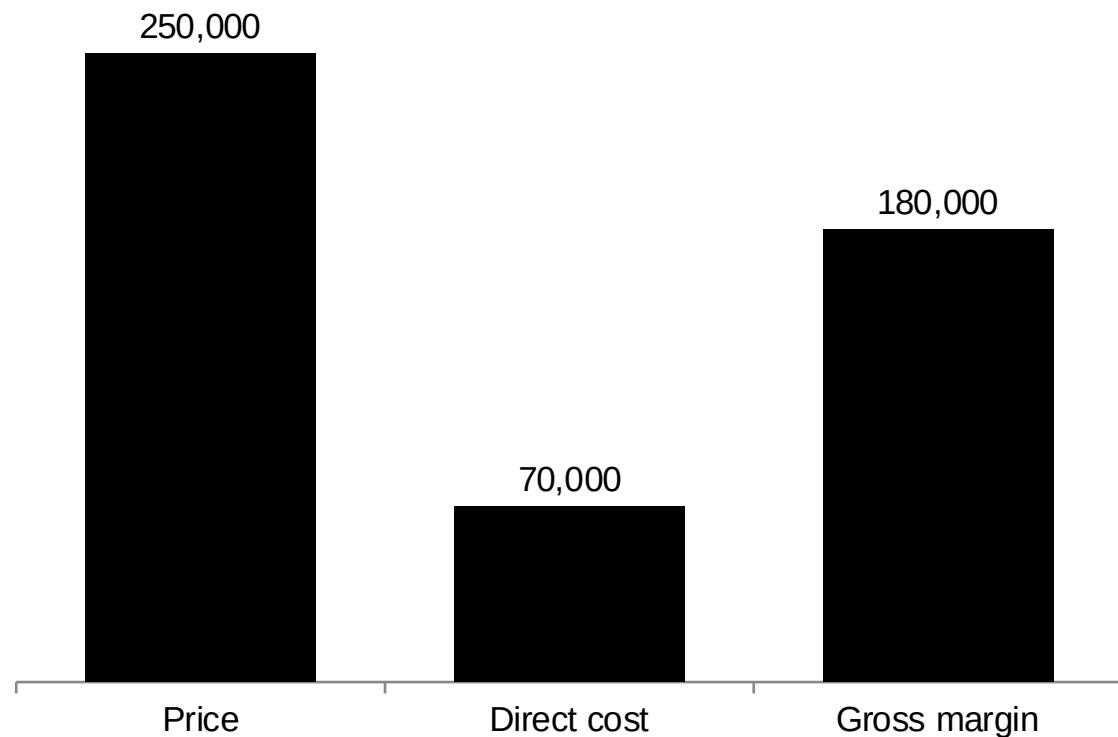
## named buyers within a day's drive

The regional catalog lists 28 quarries and sand-and-stone producers around Almaty. Asphalt and concrete plants with bulk yards are the same sale. The buyer is private, no tenders, one person says yes.

**2027.** Second vertical on the same stack, construction progress monitoring for Almaty developers.



# Subscription with service margins



## 72% gross margin

at a 250 000 KZT subscription, direct cost is about 70 000 KZT per client-month.

## 2 clients to break even

fixed costs are covered by the second subscriber.

## 30–60K KZT CAC

a free demo scan on the client's own yard, paid back by the first month.

## No cash gap

prepaid monthly, in tenge both ways. The only frozen cash is one drone, bought once.

# The rival is a habit, not a company

## **In-house surveyor**

The status quo. We complement him, he comes monthly, we fill the weeks between.

## **Contract firms, freelancers**

One-off surveys from ~150K KZT. No continuity, no stock dynamics between visits.

## **Propeller, DJI Terra**

International software proves the method works. Nobody packages it as a subscription for quarries in our region.

## **The empty niche is a question**

Untapped market or absent demand. Our first 15 customer interviews are designed to tell those apart.

# From reference pile to paying clients

W1–2



Buy the drone, build a reference pile, publish measurement error against manual survey.

W3–4



5 free demo scans across the call list, 10 customer interviews.

W5–8



Price set from interviews, first paid invoice, processing automated.

W9–12



2 paying subscriptions, 2 LOIs, a discrepancy case. Demo Day.

**Kill threshold set in advance.** Fewer than 2 paying organizations by November 15 and we call the hypothesis failed, with the data published either way.

## TEAM

# The team that signed a metro

### Amangeldi Bekaidar

Founder and CEO · full-time

Sales, negotiations, drone pilot on site visits.  
Based in Almaty.

### Myrzabay Beknurtas

Co-founder and CTO · full-time

Photogrammetry pipeline, measurement accuracy. Built LocalQuakeAlert (ESP32) solo.

### Bakhtiyar Merey

Communications · part-time

Design, content, customer interviews across both projects.

## 1st place

TURAQTY JOL 7.0

## Signed LOI

Almaty Metro, EcoStep pilot

## Invited

TechCrunch Disrupt 2026

Same three people behind EcoStep, with a split calendar between the two projects.



THE ASK

# **A tracker who has sold to heavy industry, pressure on our numbers, and a Demo Day deadline**

We bring the checked idea, the hardware budget and the call list. The program brings the discipline. By December 6 that adds up to 2 paying quarries or an honest dataset on why not.